

Bayesian Poisson Model

Load the data

```
library("haven")
library("dplyr")
wm <- read_sav("wm.sav")
wm <- wm %>% arrange(HH1, HH2, WM3)
hh <- read_sav("hh.sav")
df_hh <- hh %>% select(HH1, HH2, HHSEX, HHAGE)
wm <- merge(x=wm, y=df_hh, by = c("HH1", "HH2"))
```

Clean the data

(one line of code)

```
wm <- wm %>% filter(MA1 == 1 & MT1!=9 & MT2!=9 & MT3!=9 & MA2!=98 & MA2!=99 & MA3!=9)
```

68709 women before cleaning the data

MA1: "Currently married"

```
table(wm$MA1,useNA="ifany")
```

```
##
##      1
## 51331
```

```
wm <- wm %>% filter(MA1==1)
```

```
1      3  <NA>
51426 12951 4332
```

We exclude 12951 women who have never married and 4332 women with unknown status. In total, 17283 samples were excluded.

Technology usage (22): Missing data: media_newspaper (5), media_radio (3), media_tv(14)

```
table(wm$MT1,useNA="ifany")
```

```
##
##      0      1      2      3
## 48411 1364   884   672
```

```
table(wm$MT2,useNA="ifany")
```

```
##
##      0      1      2      3
## 50608   316   271   136
```

```
table(wm$MT3,useNA="ifany")
```

```
##
##      0      1      2      3
## 18295 2321 4162 26553

attr(wm$MT1, "labels")

##          NOT AT ALL LESS THAN ONCE A WEEK  AT LEAST ONCE A WEEK          ALMOST EVERY DAY
##                  0                      1                      2                      3
##          NO RESPONSE
##                  9

wm <- wm %>% filter(MT1!=9 & MT2!=9 & MT3!=9)

other wives: (30)

table(wm$MA3,useNA="ifany")

##
##      1      2
## 1618 49713

wm <- wm %>% filter(MA3!=9)

Age of husband (44) (98 for "DK"; 99 for "NO RESPONSE")

table(wm$MA2,useNA="ifany")

##
##  15  16  17  18  19  20  21  22  23  24  25  26  27  28  29  30  31
##   1   3  14  73  95 237 302 479 527 742 1038 1042 1055 1448 1201 2178 1495
##  32  33  34  35  36  37  38  39  40  41  42  43  44  45  46  47  48
## 1909 1456 1556 2558 1828 1567 2035 1588 2212 1485 1559 1162 1148 1822 1375 1368 1620
##  49  50  51  52  53  54  55  56  57  58  59  60  61  62  63  64  65
## 1137 1429 1029 978 819 715 1051 801 553 532 343 544 246 199 150 105 164
##  66  67  68  69  70  71  72  73  74  75  76  77  78  79  80  82  84
##  72  41  51  19  59  22  16   8   9  21  10   5   6   7   2   3   1
##  85  88  90
##   2   1   3

wm <- wm %>% filter(MA2!=98 & MA2!=99)

HHAGE

table(wm$HHAGE,useNA="ifany")

##
##  16  17  18  19  20  21  22  23  24  25  26  27  28  29  30  31  32
##   3   7  28  40  89 154 231 258 383 590 624 674 934 846 1470 1142 1327
##  33  34  35  36  37  38  39  40  41  42  43  44  45  46  47  48  49
## 1158 1311 1863 1588 1427 1732 1491 1799 1445 1399 1146 1113 1640 1376 1435 1633 1212
##   50  51  52  53  54  55  56  57  58  59  60  61  62  63  64  65  66
## 1450 1198 1078 984 902 1320 1108 810 835 705 1095 611 606 511 363 649 379
##   67  68  69  70  71  72  73  74  75  76  77  78  79  80  81  82  83
##  308 364 218 499 257 212 145  94 211 128  95 102  61 137  55  42  33
##   84  85  86  87  88  89  90  92  93  94  95
##   21  56  26  27  10   9  19   9   1   2  18
```

No missing values for the following predictors:

```
table(wm$HH6,useNA="ifany")
```

```
##
##      1      2
## 10123 41208
```

```
table(wm$HH7,useNA="ifany")
```

```
##
##     10     20     30     40     45     50     55     60
## 4539  9135 10364  8410  2670  6363  6430  3420
```

```
table(wm$HH7A,useNA="ifany")
```

```
##
##      1      3      4      6      9     10     12     13     15     18     19     22     26     27     29     30     32
## 804  600  770  726  749  790  765  677 1675  876 1094  686 1963  804  720  816  707
##  33   35   36   38   39   41   42   44   46   47   48   49   50   51   52   54   55
## 644  642  878  779  588  882  707  861  642  755  629  808  899  695  819  618  817
##  56   57   58   59   61   64   65   67   68   69   70   72   73   75   76   77   78
## 696  885  874  782  844  769  775  811  756  754  835  657  846  839  816  806  847
##  79   81   82   84   85   86   87   88   89   90   91   93   94
## 740  802  746  646  739  657  856  818  581  831  837  700  901
```

```
#table(wm$sumweight,useNA="ifany")
```

```
table(wm$WAGEM,useNA="ifany")
```

```
##
##      5      6      7      8      9     10     11     12     13     14     15     16     17     18     19     20     21
##      1      2      1      5     16     40    614 1730 3813 5751 7125 7463 6783 5599 3721 2647 1764
##     22     23     24     25     26     27     28     29     30     31     32     33     34     35     36     37     38
## 1252  889  559  473  332  221  169  105   77   56   29   35   15    8    9    8    3
##     39     41     42     43     44     45     48
##      4      4      3      1      2      1      1
```

```
table(wm$WB4,useNA="ifany")
```

```
##
##     15     16     17     18     19     20     21     22     23     24     25     26     27     28     29     30     31
## 183  356  450 1359 1483 1580 1509 1669 1707 1745 1838 1955 1753 1858 1762 2205 2015
##    32    33    34    35    36    37    38    39    40    41    42    43    44    45    46    47    48
## 1900 1848 1803 1799 2045 1630 1777 1480 1433 1405 1293 1172 1084 1021 1190 1135 1153
##    49
##   736
```

```
table(wm$windex5,useNA="ifany")
```

```
##
##      1      2      3      4      5
## 10718 10732 10702 10336  8843
```

```
table(wm$welevel,useNA="ifany")
```

```
##
##      0      1      2      3
## 9186 13235 22289  6621
```

```
table(wm$ethnicity,useNA="ifany")
```

```
##
##      1      2
## 50184 1147
```

```
table(wm$CM11,useNA="ifany")
```

```
##
##      0      1      2      3      4      5      6      7      8      9     10     11     12
## 4663 10933 16565 10670 5029 2029 885 337 145 52 18 3 2
```

Summary

```
women_age <- wm$WB4
women_age2 <- women_age^2
husband_age <- wm$MA2
husband_age2 <- husband_age^2
HH_age <- wm$HHAGE
HH_age2 <- wm$HHAGE^2
women_agem <- wm$WAGEM
division <- wm$HH7
ethnicity <- wm$ethnicity # Fixed
area <- wm$HH6 # Fixed
women_edu <- wm$welevel
HH_sex <- wm$HHSEX
other_wives <- wm$MA3
windex <- wm$windex5
media_newspaper <- ifelse(wm$MT1>0, 1, 0)
media_radio <- ifelse(wm$MT2>0, 1, 0)
media_tv <- ifelse(wm$MT3>0, 1, 0)
media <- media_newspaper|media_radio|media_tv
media <- ifelse(media,1,0)
# Response variable
CEB <- wm$CM11
# District
district <- wm$HH7A
# Sampling weight
wgt <- wm$wmweight
```

```
mean(women_age)
```

```
## [1] 32.02186
```

```
sd(women_age)
```

```
## [1] 8.701892
```

```
mean(husband_age)
```

```
## [1] 39.94711
```

```
sd(husband_age)
```

```
## [1] 10.29503
```

```
mean(HH_age)
```

```
## [1] 45.12719
```

```

sd(HH_age)

## [1] 12.78071
mean(women_agem)

## [1] 16.85521
sd(women_agem)

## [1] 3.300367
total <- dim(wm)[1]
table(division)

## division
##      10      20      30      40      45      50      55      60
## 4539  9135 10364  8410  2670  6363  6430  3420
round(table(division)/total,4)

## division
##      10      20      30      40      45      50      55      60
## 0.0884 0.1780 0.2019 0.1638 0.0520 0.1240 0.1253 0.0666
table(area)

## area
##      1      2
## 10123 41208
round(table(area)/total,4)

## area
##      1      2
## 0.1972 0.8028
attr(wm$welevel, "labels")

## Pre-primary or none      Primary      Secondary      Higher secondary+
##              0              1              2              3
##      Missing/DK
##              9
table(women_edu)

## women_edu
##      0      1      2      3
## 9186 13235 22289 6621
round(table(women_edu)/total,4)

## women_edu
##      0      1      2      3
## 0.1790 0.2578 0.4342 0.1290
attr(wm$welevel, "labels")

## Pre-primary or none      Primary      Secondary      Higher secondary+
##              0              1              2              3
##      Missing/DK
##              9

```

```

table(women_edu)

## women_edu
##      0      1      2      3
##  9186 13235 22289  6621
round(table(women_edu)/total,4)

## women_edu
##      0      1      2      3
## 0.1790 0.2578 0.4342 0.1290
attr(wm$HHSEX, "labels")

##      Male  Female Missing
##          1      2      9
table(wm$HHSEX)

##
##      1      2
## 47345  3986
round(table(wm$HHSEX)/total,4)

##
##      1      2
## 0.9223 0.0777
attr(other_wives, "labels")

##          YES          NO NO RESPONSE
##          1          2          9
table(other_wives)

## other_wives
##      1      2
## 1618 49713
round(table(other_wives)/total,4)

## other_wives
##      1      2
## 0.0315 0.9685
attr(windex, "labels")

## Poorest  Second  Middle  Fourth  Richest
##      1      2      3      4      5
table(windex)

## windex
##      1      2      3      4      5
## 10718 10732 10702 10336  8843
round(table(windex)/total,4)

## windex
##      1      2      3      4      5
## 0.2088 0.2091 0.2085 0.2014 0.1723

```

```

table(media)

## media
##      0      1
## 17834 33497

round(table(media)/total,4)

## media
##      0      1
## 0.3474 0.6526

attr(wm$ethnicity, "labels")

##      Bengali      Other Missing/DK
##          1          2          9

table(ethnicity)

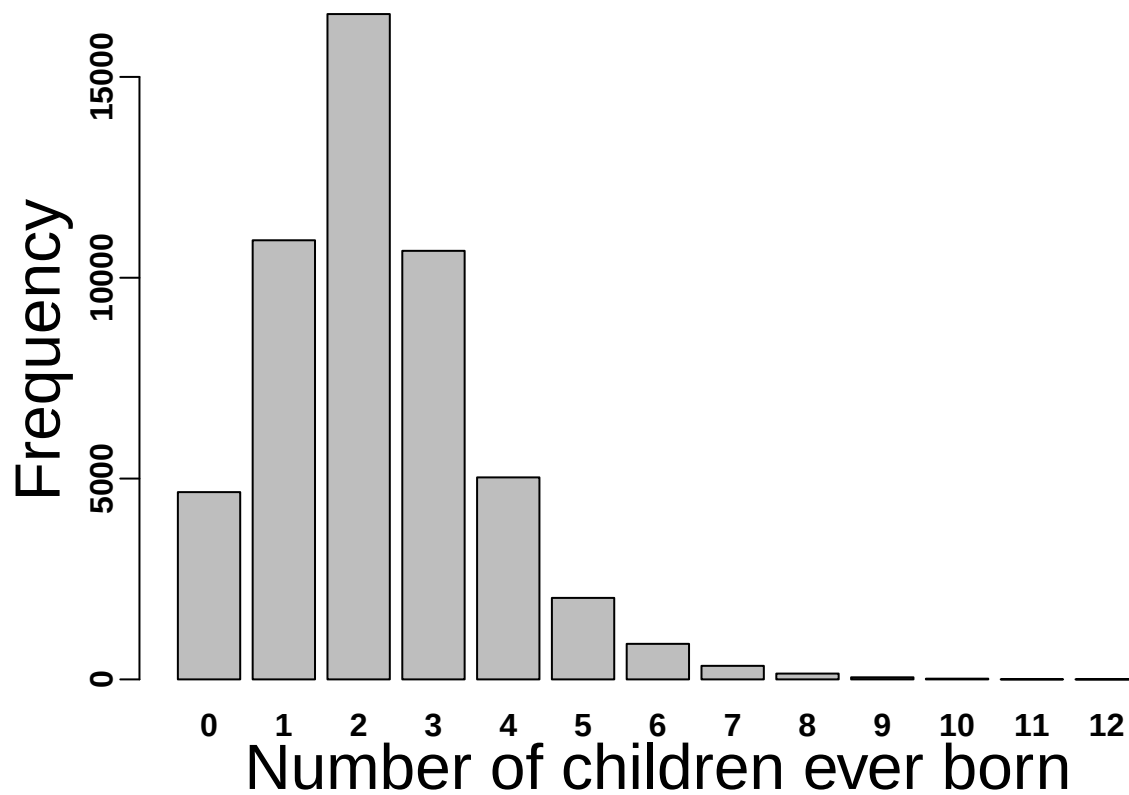
## ethnicity
##      1      2
## 50184 1147

round(table(ethnicity)/total,4)

## ethnicity
##      1      2
## 0.9777 0.0223

IMAGEOUT <- "./pics/BP/"
par(mar=c(4,4.5,1,1),mgp=c(1.88,0.50,0))
barplot(table(CEB), ylab="Frequency",
         xlab="Number of children ever born", cex.lab = 2, names.arg=c(0:12), font.axis = 2.5)

```



```
dev.print(pdf,  
    paste(IMAGEOUT,"CEB_histo.pdf",sep=""),  
    width=16,height=12)
```

```
## pdf
## 2
```

```
y <- CEB  
division <- factor(distribution, levels = c(10,20,30,40,45,50,55,60), labels = c("Barishal", "Chattogram",  
division <- relevel(distribution, ref=3)
```

```
division <- model.matrix(~division, data = division)
colnames(division) <- c(
  "Dhaka.ref",
  "Barishal",
  "Chattogram",
  "Khulna",
  "Mymensingh",
  "Rajshahi",
  "Rangpur",
  "Sylhet"
)
```

```
district_names <- c("Bagerhat", "Bandarban", "Barguna", "Barishal", "Bhola", "Bogura", "Brahmanbaria",
district_codes <- c(1, 3, 4, 6, 9, 10, 12, 13, 15, 18, 19, 22, 26, 27, 29, 30, 32, 33, 35, 36, 38, 39,
```

```
district <- factor(district, levels = district_codes, labels = district_names)
```

```
women_edu <- factor(women_edu)
women_edu <- model.matrix(~women_edu, data = women_edu)
```

```

windex <- factor(windex)
windex <- relevel(windex, ref=3)
windex <- model.matrix(~windex, data = windex)
colnames(windex) <- c(
  "windex.Middle.ref",
  "windex.Poorest",
  "windex.Poor",
  "windex.Rich",
  "windex.Richest"
)

area <- ifelse(area==2, 0, 1)
area <- factor(area)
area <- relevel(area, ref = 1)

other_wives <- ifelse(other_wives==2, 0, 1)
other_wives <- factor(other_wives)

media <- factor(media)
media <- relevel(media, ref = 1)

```

Utility functions and data

```

#####utility functions#####
## mvnrmal simulation
rmvnorm<-function(n,mu,Sigma)
{
  E<-matrix(rnorm(n*length(mu)),n,length(mu))
  t( t(E%*%chol(Sigma)) +c(mu))
}

## Wishart simulation
rwish<-function(n,nu0,S0)
{
  sS0 <- chol(S0)
  S<-array( dim=c( dim(S0),n ) )
  for(i in 1:n)
  {
    Z <- matrix(rnorm(nu0 * dim(S0)[1]), nu0, dim(S0)[1]) %*% sS0
    S[,i]<- t(Z)%*%Z
  }
  S[,1:n]
}

## mvnrm log density
ldmvnorm<-function(X,mu,Sigma,iSigma=solve(Sigma),dSigma=det(Sigma))
{
  Y<-t( t(X)-mu)
  sum(diag(-.5*t(Y)%*%Y%*%iSigma)) -
  .5*( prod(dim(X))*log(2*pi) + dim(X)[1]*log(dSigma) )
}

## Calculate matrix multiplication: (1,Z) %*% Beta
matmul <- function(Z, Beta){

```

```

nZ <- nrow(Z)
as.matrix(cbind(rep(1,nZ),Z))%*%as.matrix(Beta)
}
#####utility functions#####

#####design matrix#####
# Design matrix for all variables (random x14 | fixed x21)
X <- as.matrix(cbind(
  ### random predictors
  women_age,
  women_age2,
  husband_age,
  #husband_age2,
  HH_age,
  #HH_age2,
  women_agem,
  women_edu[,-1],
  windex[,-1],
  media,
  other_wives,
  area,
  HH_sex,
  ### fixed predictor
  division[,-1],# 7
  ethnicity
))

# Concat response variable y with design matrix of X
yX <- cbind(y,X)
#####design matrix#####

```

Init

```

#####initial values#####
p<-dim(X)[2] + 1# number of columns in X (20) + 1 (intercept)

# The initial mean of the MVN prior of theta
model <- glm(y~., data = data.frame(yX), family=poisson)
THETA <- mu0 <- model$coefficients

# The initial variance of the MVN prior for the variance of theta
# LAMBDA is the inverse of Fisher information matrix
LAMBDA <- vcov(model)

# The accepted count of proposed theta
ACCEPT.Theta.count <- 0
SIGMA.post<-NULL
THETA.post <- list()
S=220000
B=20000
thin=50
#####initial values#####

```

```
#####simulation#####
set.seed(123)
start.time <- Sys.time()
for(s in 1:S)
{
  ##update theta
  # theta proposed
  K_theta <- 0.05
  theta.p<-t(rmvnorm(1,THETA,K_theta*LAMBDA))
  lr<-0
  lr <- lr+sum(
    wgt * dpois(y,
      lambda=exp(matmul(X,theta.p)),log=TRUE),
    -wgt * dpois(y,
      lambda=exp(matmul(X,THETA)),log=TRUE),
    ldmvnorm(t(theta.p),mu0,LAMBDA),
    -ldmvnorm(t(THETA),mu0,LAMBDA)
  )
  #print("lr is")
  #print(lr)
  if( log(runif(1))<lr ) {
    ACCEPT.Theta.count<- ACCEPT.Theta.count+1
    THETA<-theta.p
  }

  ##store some output
  if((s %% thin == 0) & (s > B)) # saving every 10th value
  {
    cat("Current iteration index = ", s, "\n")
    # Print current acceptance rate
    cat("Current acceptance rate of Theta = ", round((ACCEPT.Theta.count/s)*100, digits = 2),"\n")
    # save thinned coefficients
    THETA.post<-rbind(THETA.post,t(THETA))
  }
}
end.time <- Sys.time()
#####simulation#####
```

Save and load results

```
RESULTPATH = "./data_output/BP/"
save(THETA.post, file=paste(RESULTPATH,"THETA.post.S=",S,".RData",sep=""))
save(ACCEPT.Theta.count, file=paste(RESULTPATH,"ACCEPT.Theta.count.S=",S,".RData",sep=""))

RESULTPATH = "./data_output/BP/"
load(paste(RESULTPATH,"THETA.post.S=",S,".RData",sep=""))
load(paste(RESULTPATH,"ACCEPT.Theta.count.S=",S,".RData",sep=""))
```

Plot results

Assess the convergence of the Markov chain

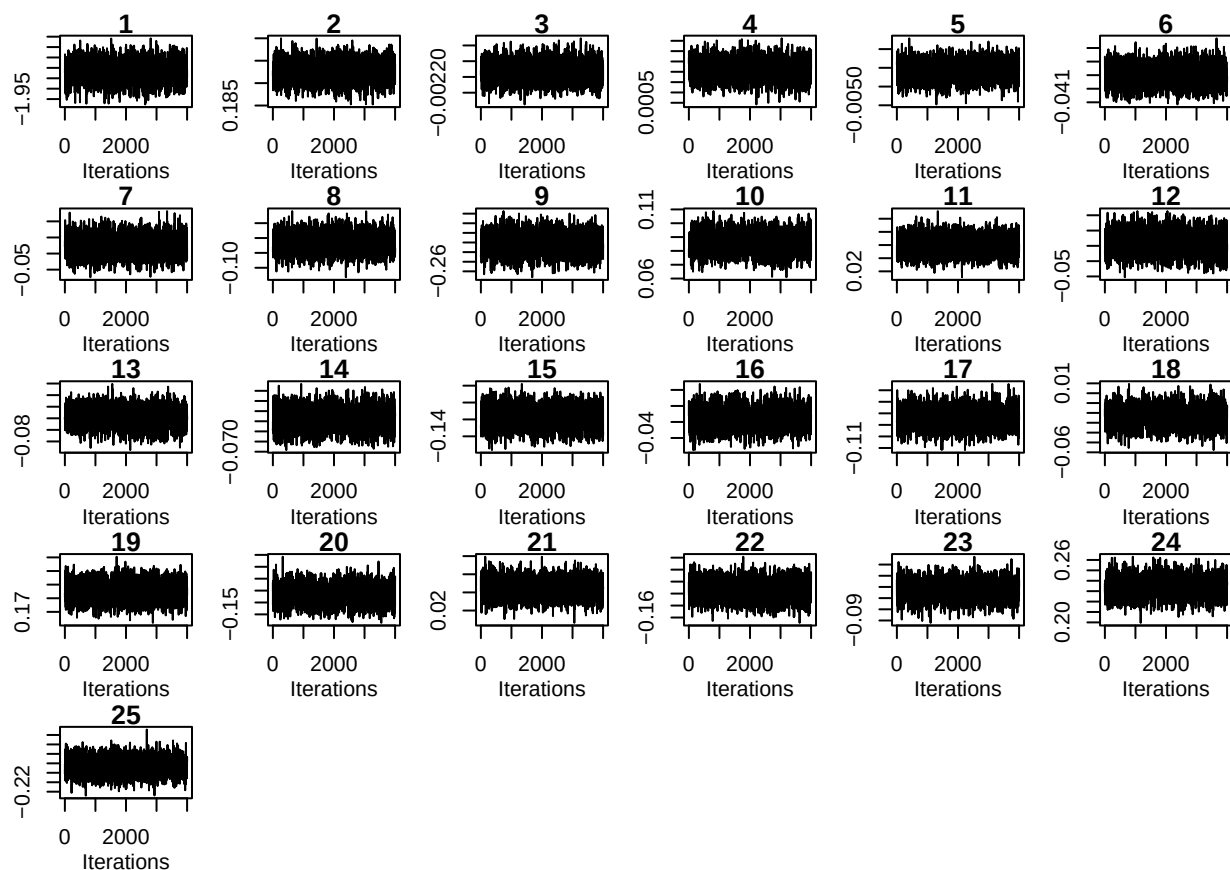
```
## stationarity plot - boxplot
stationarity.plot<-function(x,...){
  S<-length(x)
  scan<-1:S
  ng<-min( round(S/100),10)
  group<-S*ceiling( ng*scan/S) /ng
  boxplot(x~group,...)
}

# convert the result into a 26x(S-B) matrix
THETA.post.mat <- NULL
for (i in c(1:dim(THETA.post)[1])){
  THETA.post.mat <- cbind(THETA.post.mat, as.matrix(unlist(t(THETA.post)[,i])) )
}

#stationarity.plot(THETA.post[, 1],xlab="iteration",ylab=expression(theta[11]))
par(mfrow = c(5, 6), mar=c(3,2,1,1),mgp=c(2,1,0))
for (i in 1:p) {
  plot(seq(1,(S-B)/thin,by=1), THETA.post[,i], type = "l", xlab = "Iterations", ylab = expression(theta),
  )
  dev.print(pdf,
    paste(IMAGEOUT,"THETA.post.pdf",sep=""),
    width=16,height=12)
}

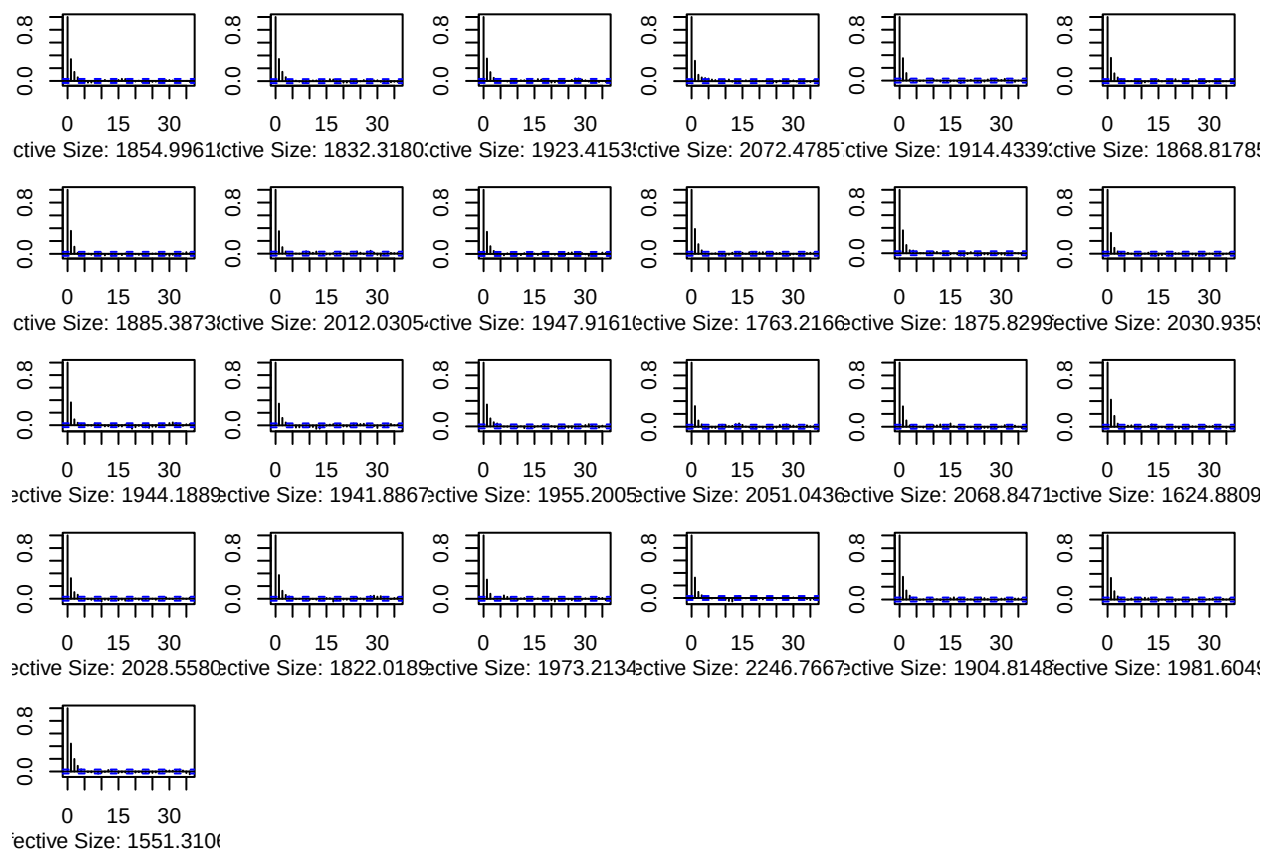
## pdf
## 2

library(coda)
par(mfrow = c(5, 6), mar=c(3,2,1,1),mgp=c(2,1,0))
```



```
for (i in 1:p){
  esize <- effectiveSize(unlist(THETA.post[,i]))
  acf_plot <- acf(unlist(THETA.post[,i]), plot = FALSE)
  t <- paste(i,"Effective Size: ", esize, sep="")
  plot(acf_plot, xlab=t, ylab="", main=t)
}
dev.print(pdf,
  paste(IMAGEOUT,"ACF.THETA.post.pdf",sep=""),
  width=16,height=12)
```

```
## pdf
## 2
```



Dispersion Test

```
fit.mle <- glm(y~., data = data.frame(yX), family=poisson)
library(AER)
```

```
## Loading required package: car
## Loading required package: carData
##
## Attaching package: 'car'
## The following object is masked from 'package:dplyr':
##
##   recode
## Loading required package: lmtest
## Loading required package: zoo
##
## Attaching package: 'zoo'
## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric
## Loading required package: sandwich
## Loading required package: survival
```

```
sink(paste(IMAGEOUT,"dispersion.txt",sep=""))
dispersiontest(fit.mle, alternative = "less")
sink()
```

Posterior estimates with 95% credible intervals (CI) of the incidence rate ratio (IRR) for the explanatory variables

```
library(xtable)
CI_THETA <- round(t(apply(exp(THETA.post.mat), 1, quantile ,probs=c(0.50, 0.025, 0.975)))), digits = 5)
CI_THETA
```

```
##           50%    2.5%   97.5%
## (Intercept) 0.16220 0.14750 0.17789
## women_age   1.21246 1.20718 1.21755
## women_age2  0.99787 0.99781 0.99793
## husband_age 1.00209 1.00121 1.00300
## HH_age      0.99592 0.99546 0.99639
## women_agem  0.96180 0.96041 0.96308
## women_edu1  0.96594 0.95482 0.97698
## women_edu2  0.92008 0.90933 0.93144
## women_edu3  0.79339 0.77776 0.80920
## windex.Poorest 1.08988 1.07509 1.10514
## windex.Poor  1.04069 1.02756 1.05459
## windex.Rich  0.97236 0.95964 0.98558
## windex.Richest 0.94285 0.92834 0.95693
## media       0.94748 0.93843 0.95679
## other_wives  0.88892 0.86870 0.90911
## area        0.97235 0.96119 0.98399
## HH_sex      0.91922 0.90520 0.93430
## Barishal    0.97670 0.95871 0.99464
## Chattogram  1.20334 1.18802 1.21920
## Khulna      0.87529 0.86237 0.88816
## Mymensingh  1.04571 1.02589 1.06578
## Rajshahi    0.87209 0.85901 0.88512
## Rangpur     0.93905 0.92461 0.95340
## Sylhet      1.26292 1.24173 1.28503
## ethnicity   0.84690 0.81909 0.87628
```

```
sink(paste(IMAGEOUT,"CI_THETA.tex",sep=""))
xtable(CI_THETA, digits = 3)
sink()
```

```
# Calculation of BIC (Weighted)
```

```
n<-length(y)
BIC_W <- p * log(n) - 2 * sum(wgt*dpois(y,exp(matmul(X,as.matrix(apply(THETA.post.mat, 1, mean))))),log=
print(BIC_W)
```

```
## [1] 149880.5
```

Calculate the expected number of CEB

Expected Children Ever Born - Woman's age

```
Xs <- apply(X, 2, mean)
Xs <- matrix(rep(Xs, 35), nrow = 35, byrow = T)
```

```

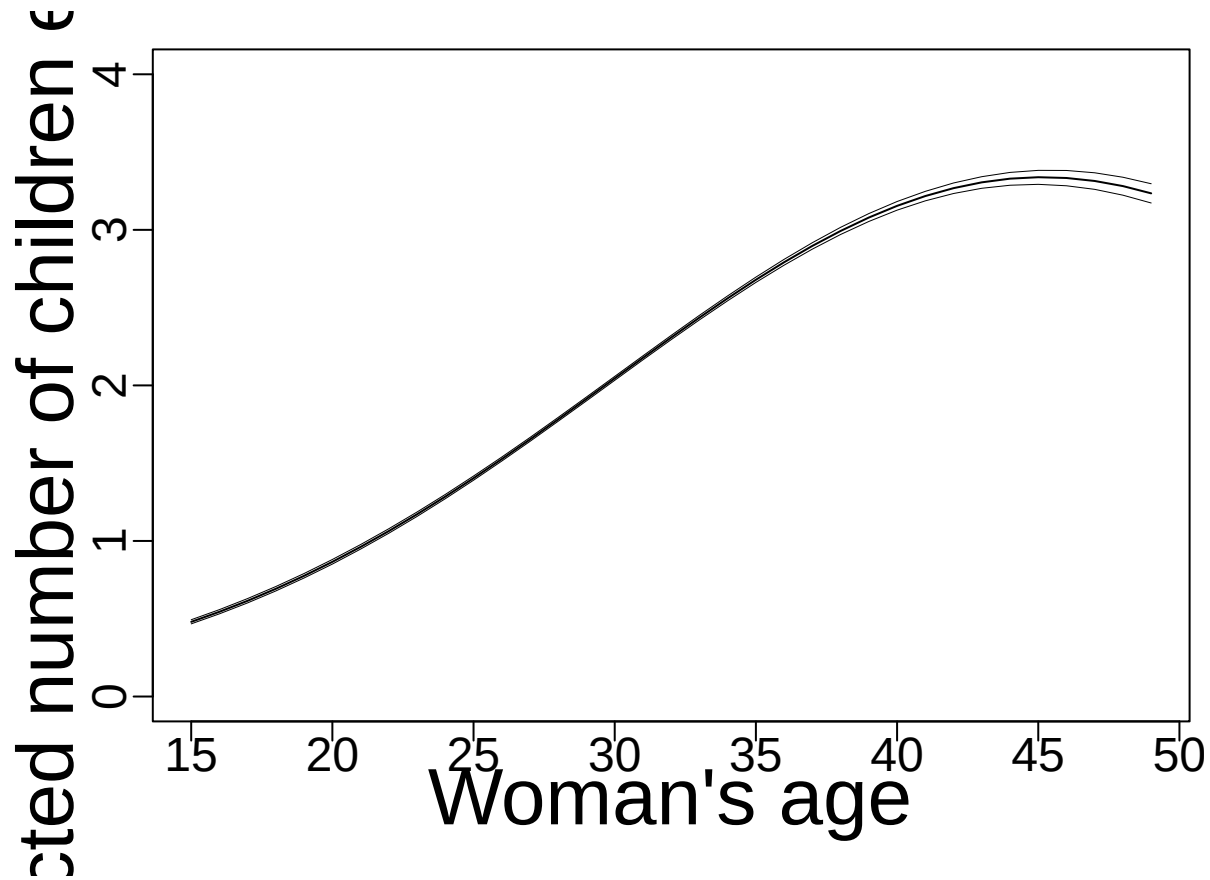
Xs[,1] <- seq(15, 49, by = 1)
Xs[,2] <- (seq(15, 49, by = 1)^2)
colnames(Xs) <- colnames(X)
Xs = cbind(rep(1,35),Xs) # insert the first column of 1 for intercept
eXB.post <- exp(Xs*%THETA.post.mat)
qE<-apply(t(eXB.post),2,quantile,probs=c(.025,.5,.975))

sample_size <- NULL
for (v in 15:49){
  sample_size <- c(sample_size, sum(X[,1]==v))
}
qE <- rbind(qE,sample_size)

colnames(qE)<-as.character(15:49)
write.csv(t(qE), paste(IMAGEOUT,"qE.women_age.csv",sep=""))

par(mar=c(4,4.5,1,1),mgp=c(1.88,0.50,0))
plot( c(15, 49),range(c(0,qE)),type="n",xlab="Woman's age",ylim=c(0,4),
      ylab="Expected number of children ever born", cex.lab = 2.5, cex.axis = 1.5)
lines(seq(15, 49, by = 1), qE[1,],col="black",lwd=0.5)
lines(seq(15, 49, by = 1), qE[2,],col="black",lwd=1.0)
lines(seq(15, 49, by = 1), qE[3,],col="black",lwd=0.5)

```



```

dev.print(pdf,
  paste(IMAGEOUT, "E_women_age.pdf", sep=""),
  width=16,height=16)

```

```
## pdf
## 2
```

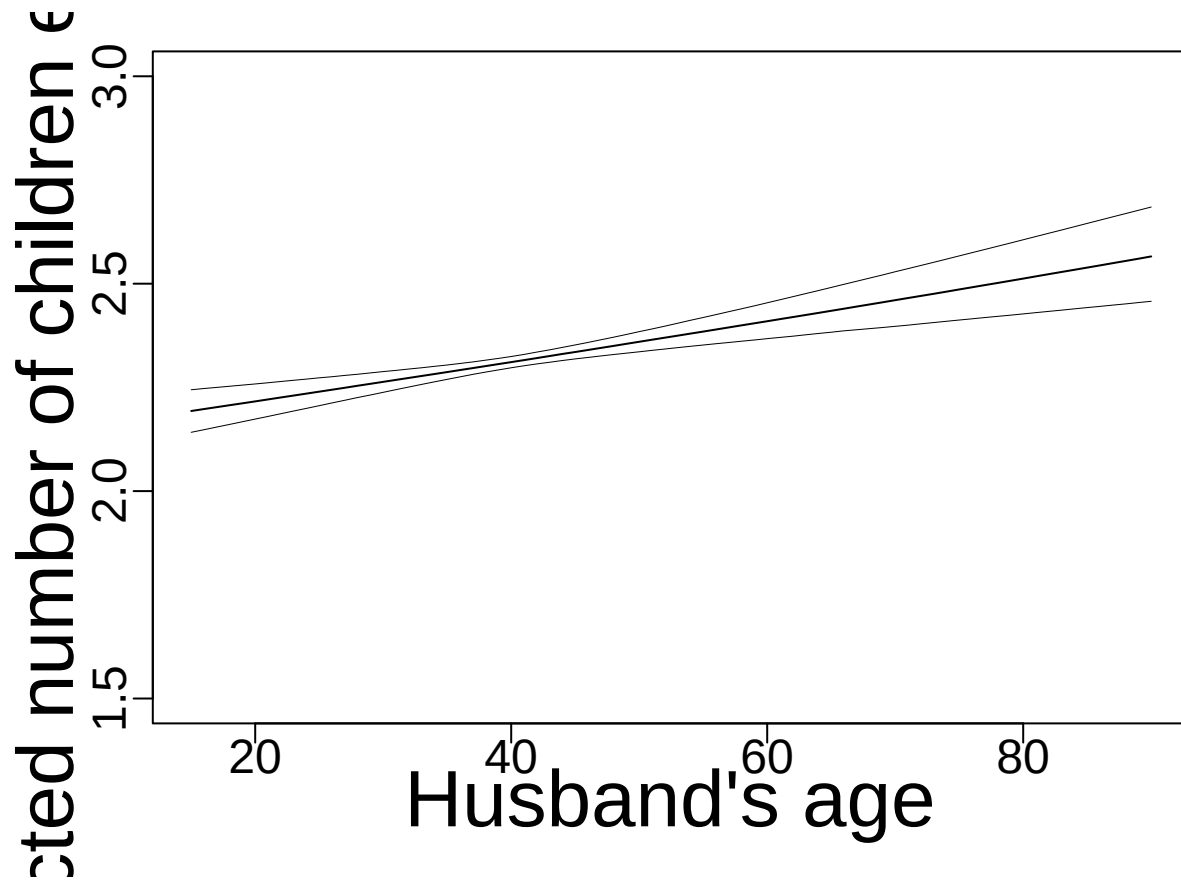
Expected Children Ever Born - Husband's age

```
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- matrix(rep(Xs, 76), nrow = 76, byrow = T)
Xs[,3] <- seq(15, 90, by = 1)
colnames(Xs) <- colnames(X)
Xs = cbind(rep(1,76),Xs) # insert the first column of 1 for intercept
eXB.post <- exp(Xs%%THETA.post.mat)
qE<-apply(t(eXB.post),2,quantile,probs=c(.025,.5,.975))

sample_size <- NULL
for (v in 15:90){
  sample_size <- c(sample_size, sum(X[,3]==v))
}
qE <- rbind(qE,sample_size)

colnames(qE)<-as.character(15:90)
write.csv(t(qE), paste(IMAGEOUT,"qE.husband_age.csv",sep=""))

par(mar=c(4,4.5,1,1),mgp=c(1.88,0.50,0))
plot( c(15, 90),range(c(0,qE)),type="n",xlab="Husband's age",ylim=c(1.5,3),
      ylab="Expected number of children ever born", cex.lab = 2.5, cex.axis = 1.5)
lines(seq(15, 90, by = 1), qE[1,],col="black",lwd=0.5)
lines(seq(15, 90, by = 1), qE[2,],col="black",lwd=1.0)
lines(seq(15, 90, by = 1), qE[3,],col="black",lwd=0.5)
```



```
dev.print(pdf,
  paste(IMAGEOUT, "E_husband_age.pdf", sep=""),
  width=16, height=16)
```

```
## pdf
## 2
```

Expected Children Ever Born - Household head's age

```
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- matrix(rep(Xs, 80), nrow = 80, byrow = T)
Xs[,4] <- seq(16, 95, by = 1)
colnames(Xs) <- colnames(X)
Xs = cbind(rep(1,80),Xs) # insert the first column of 1 for intercept
eXB.post <- exp(Xs*%THETA.post.mat)
qE<-apply(t(eXB.post),2,quantile,probs=c(.025,.5,.975))

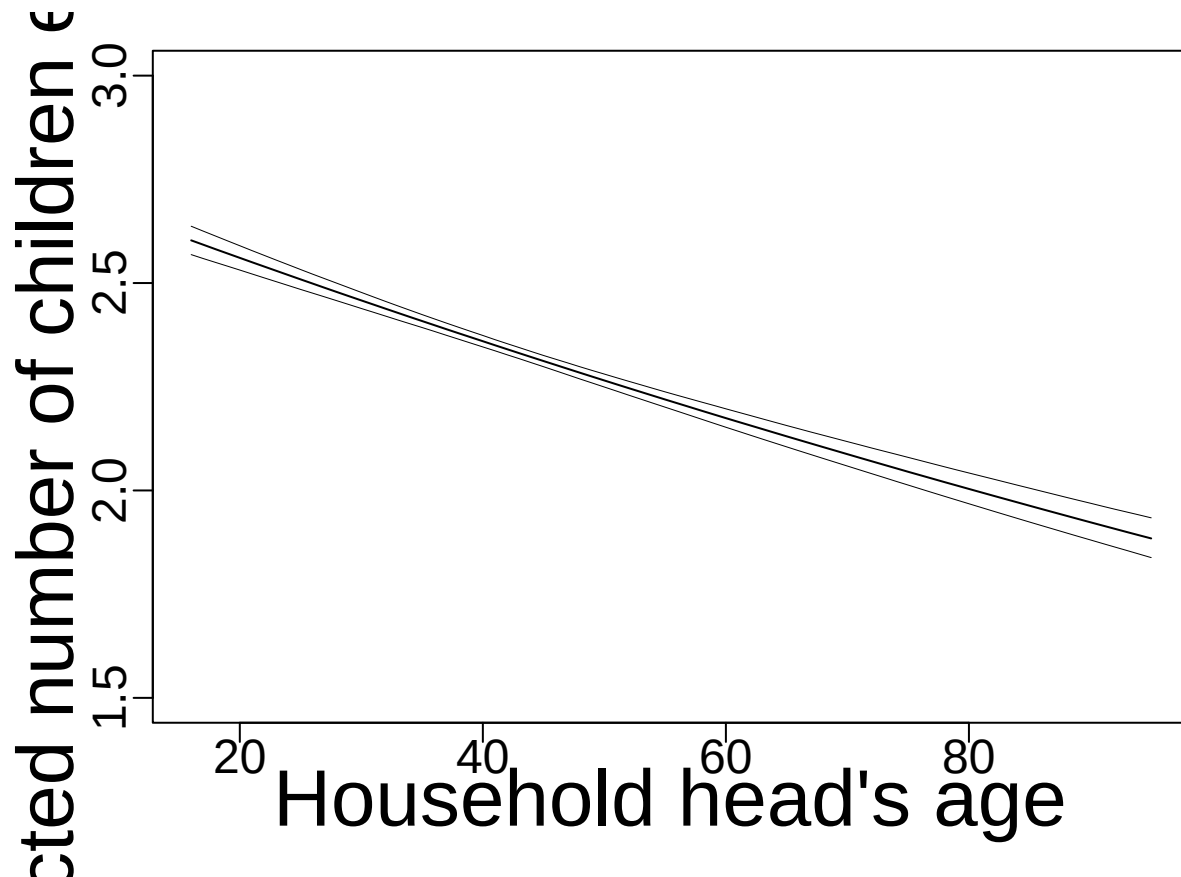
sample_size <- NULL
for (v in 16:95){
  sample_size <- c(sample_size, sum(X[,4]==v))
}
qE <- rbind(qE,sample_size)

colnames(qE)<-as.character(16:95)
write.csv(t(qE), paste(IMAGEOUT, "qE.HH_age.csv", sep=""))
```

```

par(mar=c(4,4.5,1,1),mgp=c(1.88,0.50,0))
plot( c(16, 95),range(c(0,qE)),type="n",xlab="Household head's age",ylim=c(1.5,3),
      ylab="Expected number of children ever born", cex.lab = 2.5, cex.axis = 1.5)
lines(seq(16, 95, by = 1), qE[1,],col="black",lwd=0.5)
lines(seq(16, 95, by = 1), qE[2,],col="black",lwd=1.0)
lines(seq(16, 95, by = 1), qE[3,],col="black",lwd=0.5)

```



```

dev.print(pdf,
  paste(IMAGEOUT, "E_HH_age.pdf", sep=""),
  width=16,height=16)

```

```

## pdf
## 2

```

Expected Children Ever Born - Woman's age at marriage

```

Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- matrix(rep(Xs, 44), nrow = 44, byrow = T)
Xs[,5] <- seq(5, 48, by = 1)
colnames(Xs) <- colnames(X)
Xs = cbind(rep(1,44),Xs)
eXB.post <- exp(Xs*%THETA.post.mat)
qE<-apply(t(eXB.post),2,quantile,probs=c(.025,.5,.975))

sample_size <- NULL

```

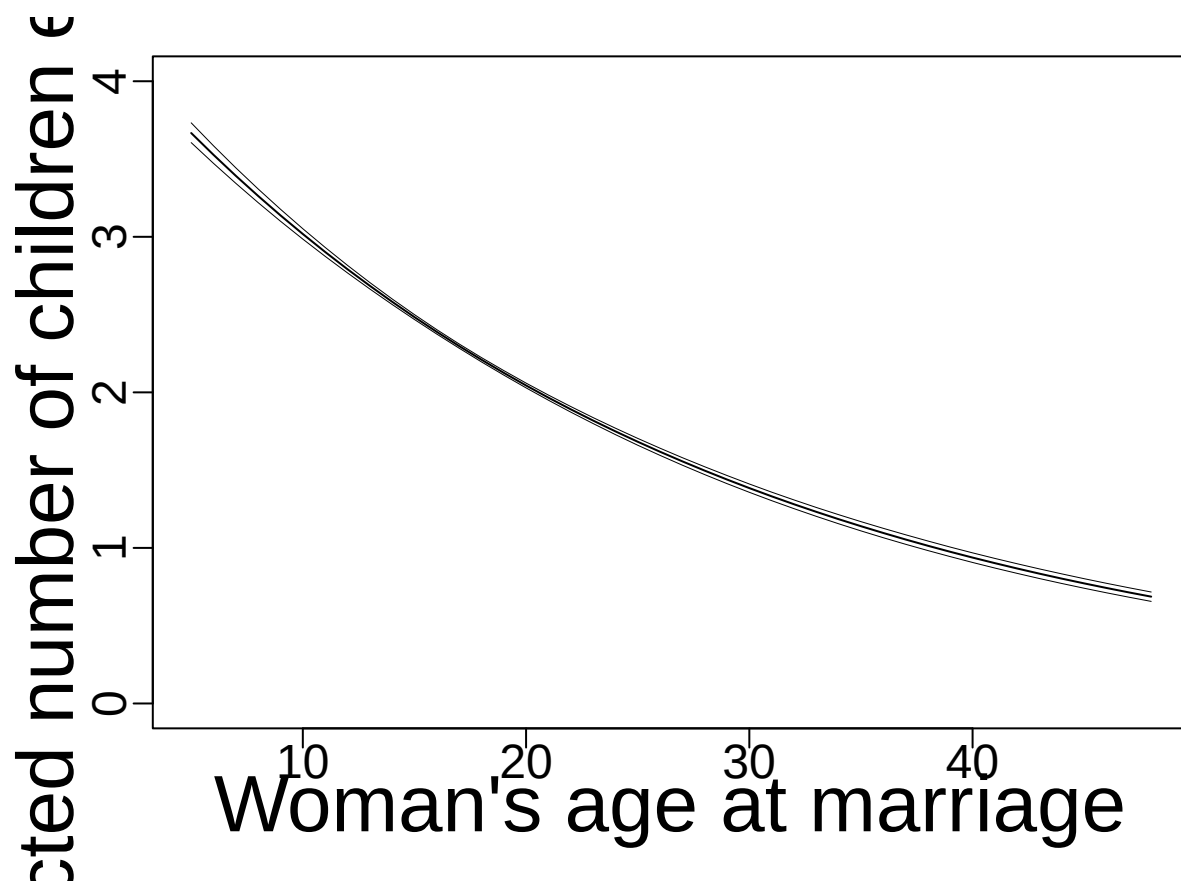
```

for (v in 5:48){
  sample_size <- c(sample_size, sum(X[,5]==v))
}
qE <- rbind(qE,sample_size)

colnames(qE)<-as.character(5:48)
write.csv(t(qE), paste(IMAGEOUT,"qE.women_agem.csv",sep=""))

par(mar=c(4,4.5,1,1),mgp=c(1.88,0.50,0))
plot( c(5, 48),range(c(0,qE)),type="n",xlab="Woman's age at marriage",ylim=c(0,4),
      ylab="Expected number of children ever born", cex.lab = 2.5, cex.axis = 1.5)
lines(seq(5, 48, by = 1), qE[1,],col="black",lwd=0.5)
lines(seq(5, 48, by = 1), qE[2,],col="black",lwd=1.0)
lines(seq(5, 48, by = 1), qE[3,],col="black",lwd=0.5)

```



```

dev.print(pdf,
  paste(IMAGEOUT,"E_women_agem.pdf",sep=""),
  width=16,height=16)

```

```

## pdf
## 2

```

Expected Children Ever Born - Division

```

# Expected Number of Children Ever Born in Bangladesh
Xs <- apply(X, 2, mean)

```

```

Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
eXB.post<- exp(t(t(Xs)%*%THETA.post.mat) )
qEA<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
round(qEA, digits = 3)

##          [,1]
## 2.5%    2.297
## 50%     2.311
## 97.5%   2.324

qEA <- rbind(qEA,dim(X)[1])
# Division

dnames <- c(
  "Bangladesh",
  "Dhaka",
  "Barishal",
  "Chattogram",
  "Khulna",
  "Mymensingh",
  "Rajshahi",
  "Rangpur",
  "Sylhet"
)

Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- matrix(rep(Xs, 8), nrow = 8, byrow = T)
for (div_index in c(2:8)){
  x <- rep(0,8)
  x[div_index] <- 1
  Xs[, (15+div_index)] <- x
}
colnames(Xs) <- colnames(X)
Xs = cbind(rep(1,8),Xs)

eXB.post<- exp(t(Xs)%*%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))

sample_size <- sum(wm$HH7==30) # count number of women in Herat(reference levels)
for (HH7 in c(10,20,40,45,50,55,60)){
  sample_size <- c(sample_size, sum(wm$HH7==HH7))
}
qE <- rbind(qE,sample_size)

qE.out <- cbind(qEA,qE)
colnames(qE.out)<-dnames
write.csv(t(qE.out), paste(IMAGEOUT,"qE.province.csv",sep=""))

index <- order(qE[2,])
index <- c(1, index + 1)

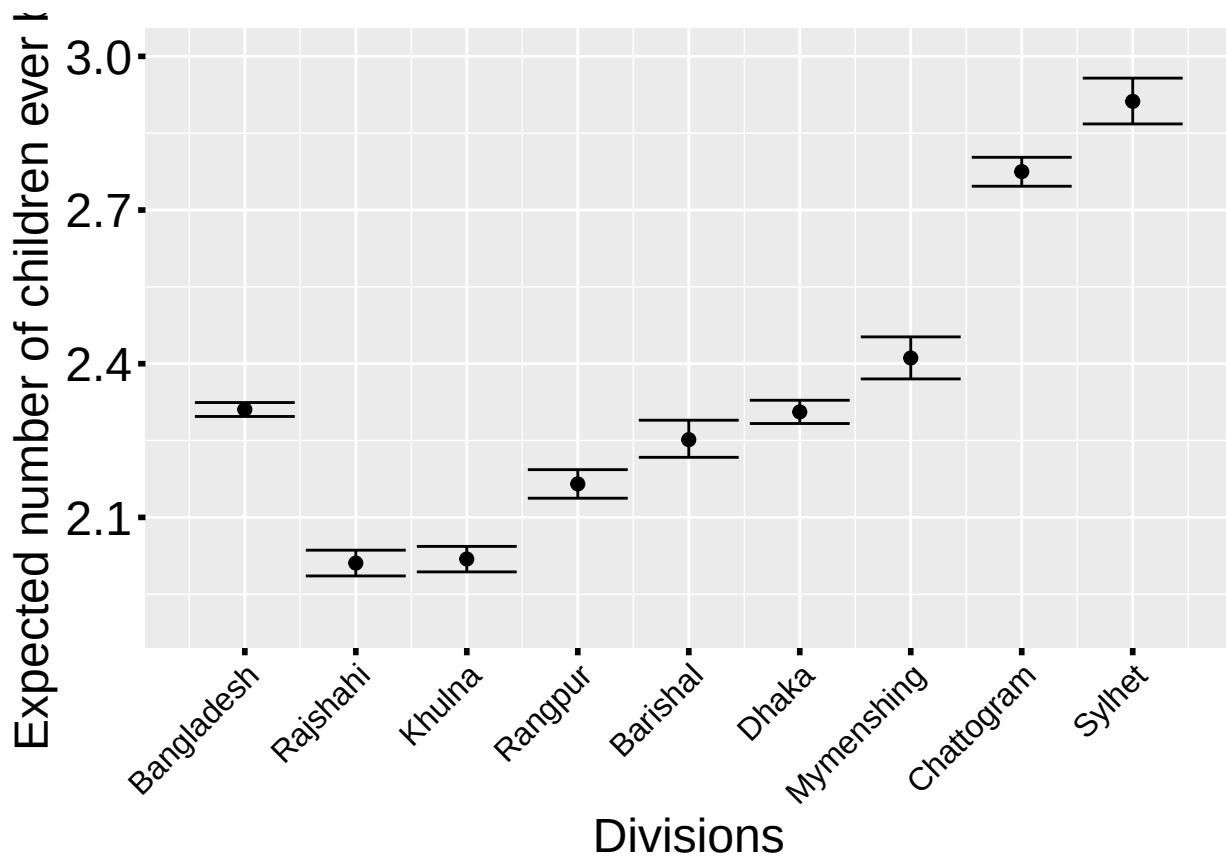
library(ggplot2)

```

```

BP.df_division <- data.frame(x =1:9,
                             F = qE.out[2,][index],
                             L = qE.out[1,][index],
                             U = qE.out[3,][index],
                             name = dnames[index])
abc <- ggplot(BP.df_division, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Divisions") +
  ylim(1.9,3.0) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:9), labels = dnames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 12, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)

```



```

save(BP.df_division, file=paste(IMAGEOUT,"BP.df_division.RData",sep=""))
#dev.print(device = postscript, paste(IMAGEOUT, "E_province.eps",sep=""),width=8,height=8, horizontal = "t")
dev.print(pdf, paste(IMAGEOUT, "E_division.pdf",sep=""),width=8,height=8)

## pdf
## 2

```

Expected Children Ever Born - Ethnicity

```
anames<- c("Bengali", "Other")
index <- c(1, 2)
anames[index]

## [1] "Bengali" "Other"

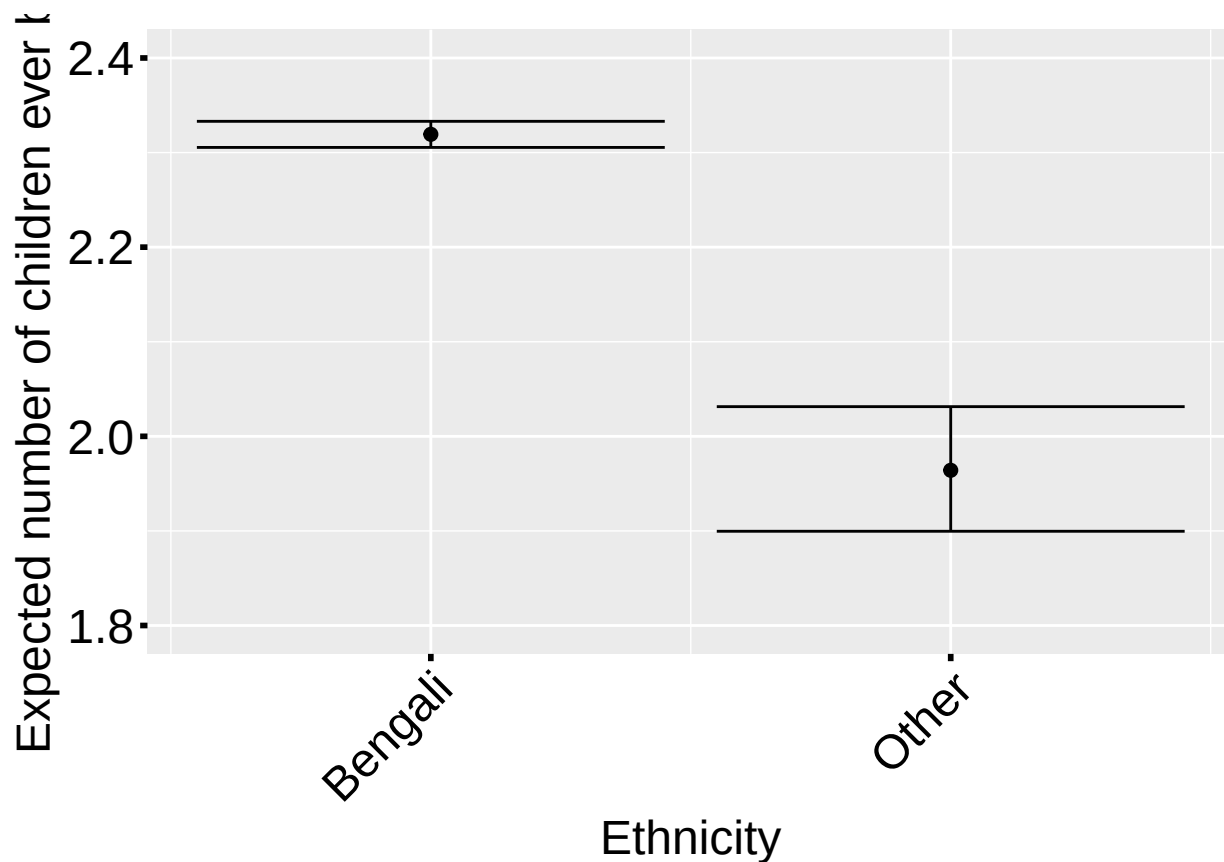
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
Xs <- matrix(rep(Xs, 2), nrow = 2, byrow = T)
Xs[,25] <- c(1, 2) # 2 for Other, 1 for Bengali

eXB.post<- exp(t(Xs%*%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))

sample_size <- c(sum(X[,23]==1), sum(X[,23]==2)) # Bengali , Other
qE <- rbind(qE,sample_size)

colnames(qE)<-anames
write.csv(t(qE), paste(IMAGEOUT,"qE.ethnicity.csv",sep=""))

BP.df_ethnicity <- data.frame(x =1:2,
                             F = c(qE[2,])[index],
                             L = c(qE[1,])[index],
                             U = c(qE[3,])[index])
abc <- ggplot(BP.df_ethnicity, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Ethnicity") +
  geom_point(size = 2) +
  ylim(1.8,2.4) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:2), labels = anames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 0),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)
```



```
save(BP.df_ethnicity, file=paste(IMAGEOUT,"BP.df_ethnicity.RData",sep=""))
dev.print(pdf, paste(IMAGEOUT, "E_ethnicity.pdf",sep=""),width=8,height=8)
```

```
## pdf
## 2
```

```
(qE[2,][1] - qE[2,][2])/qE[2,][2]
```

```
## Bengali
## 0.1808577
```

Expected Children Ever Born - Area

```
anames<- c("Rural", "Urban")
index <- c(1, 2)
anames[index]
```

```
## [1] "Rural" "Urban"
```

```
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
Xs <- matrix(rep(Xs, 2), nrow = 2, byrow = T)
Xs[,16] <- c(1, 2) # 2 for Urban, 1 for Rural
```

```
eXB.post<- exp(t(Xs*%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
```

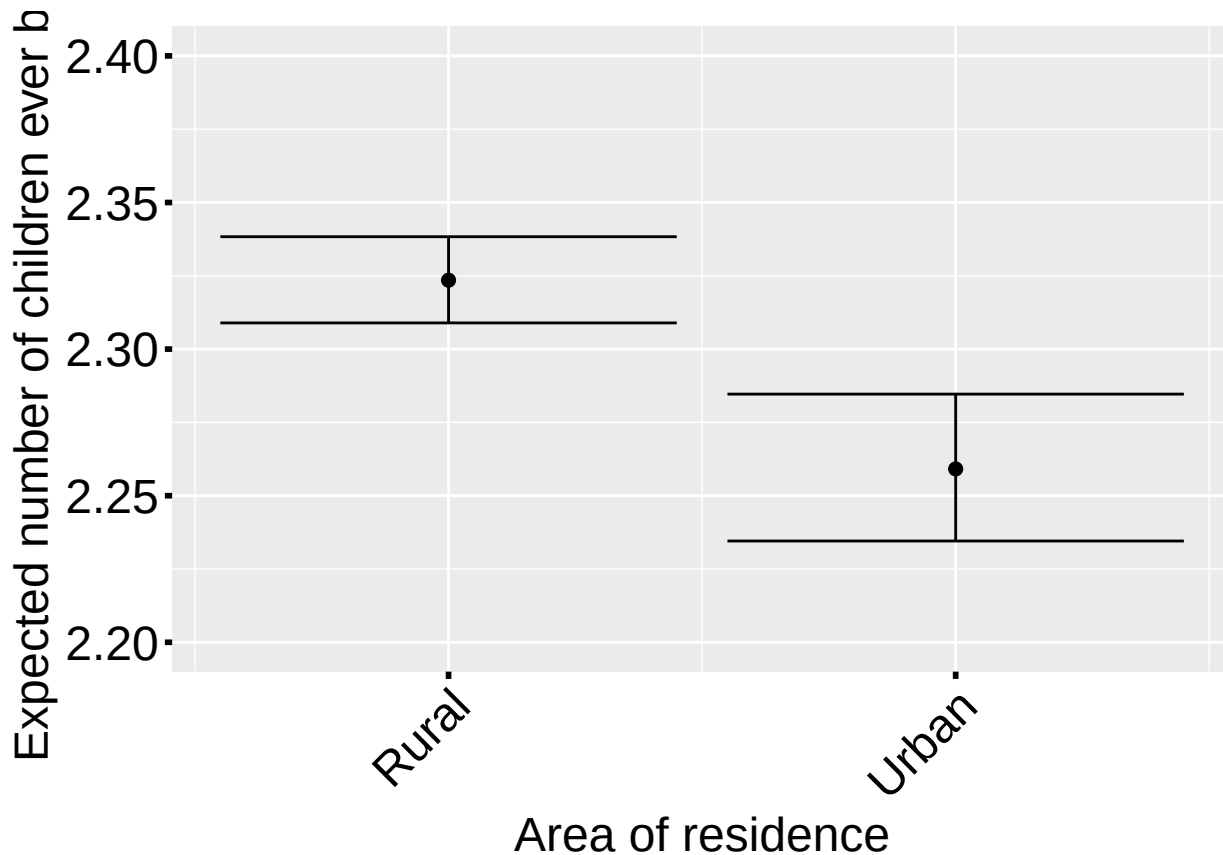
```

sample_size <- c(sum(X[,15]==1), sum(X[,15]==2)) # Rural, Urban
qE <- rbind(qE,sample_size)

colnames(qE)<-anames
write.csv(t(qE), paste(IMAGEOUT,"qE.area.csv",sep=""))

BP.df_area <- data.frame(x =1:2,
                        F = c(qE[2,])[index],
                        L = c(qE[1,])[index],
                        U = c(qE[3,])[index])
abc <- ggplot(BP.df_area, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Area of residence") +
  geom_point(size = 2) +
  ylim(2.2,2.4) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:2), labels = anames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)

```



```

save(BP.df_area, file=paste(IMAGEOUT,"BP.df_area.RData",sep=""))
dev.print(pdf, paste(IMAGEOUT, "E_area.pdf",sep=""),width=8,height=8)

```

```

## pdf
## 2

```

```
(qE[2,][1] - qE[2,][2])/qE[2,][2]
```

```
##      Rural
## 0.02849036
```

Expected Children Ever Born - Women's education

```
winames <- c("Pre-primary or none", "Primary", "Secondary", "Higher secondary+")
index <- c(1, 2, 3, 4)
winames[index]
```

```
## [1] "Pre-primary or none" "Primary"          "Secondary"
## [4] "Higher secondary+"      
```

```
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
Xs <- matrix(rep(Xs, 4), nrow = 4, byrow = T)
Xs[,7] <- c(0, 1, 0, 0)
Xs[,8] <- c(0, 0, 1, 0)
Xs[,9] <- c(0, 0, 0, 1)
```

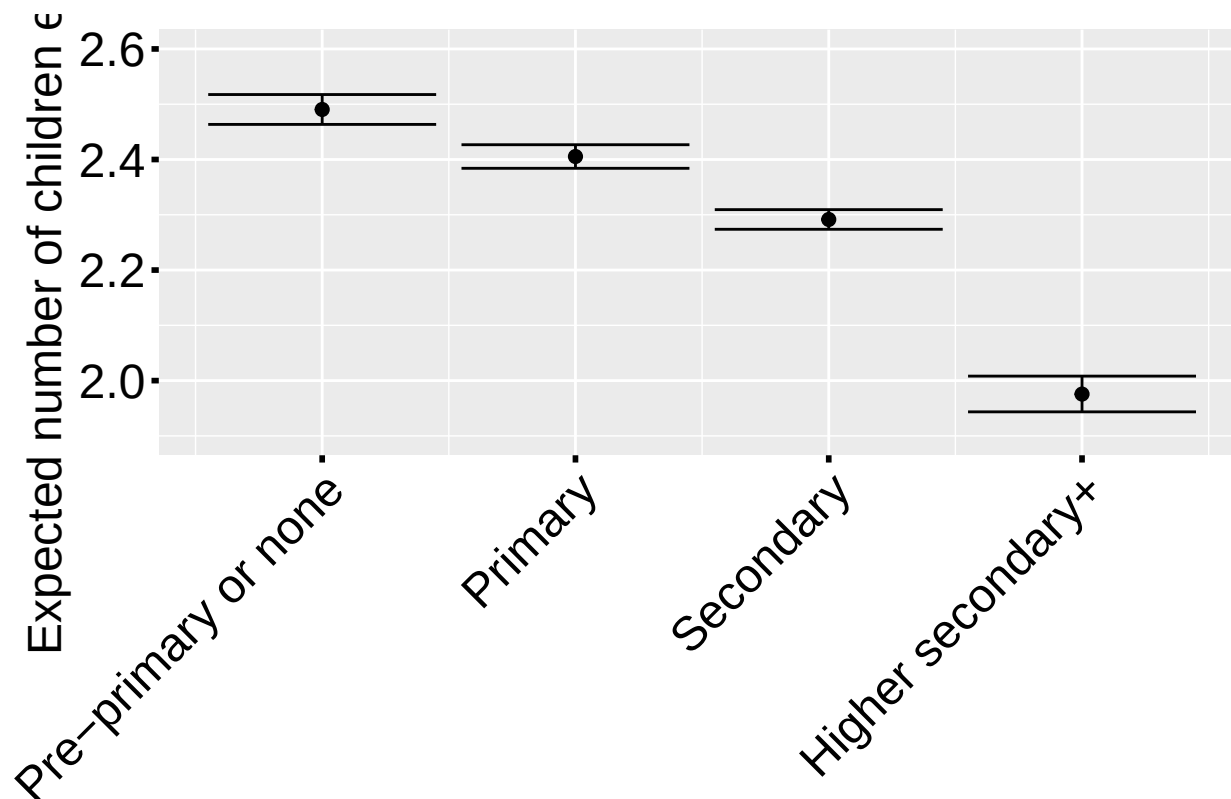
```
eXB.post<- exp(t(Xs%*%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
```

```
sample_size <- NULL
for (edu in 0:3){
  sample_size <- c(sample_size, sum(wm$welevel==edu))
}
qE <- rbind(qE,sample_size)
```

```
colnames(qE)<-winames
write.csv(t(qE), paste(IMAGEOUT,"qE.women_edu.csv",sep=""))
```

```
BP.df_women_edu <- data.frame(x =1:4,
                              F = c(qE[2,][index],
                              L = c(qE[1,][index],
                              U = c(qE[3,][index])
```

```
abc <- ggplot(BP.df_women_edu, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Women's education", cex.lab = 2.5, cex.axis =
  geom_point(size = 2) +
  ylim(1.9,2.6) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:4), labels = winames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle =
    axis.ticks.x = element_line(color = "black", linewidth = 1),
    axis.text.y = element_text(size = 18, color = "black"),
    axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)
```



Women's education

```
save(BP.df_women_edu, file=paste(IMAGEOUT, "BP.df_women_edu.RData", sep=""))

dev.print(pdf, paste(IMAGEOUT, "E_women_edu.pdf", sep=""), width=8, height=8)

## pdf
## 2
```

Expected Children Ever Born - Wealth Index

```
winames <- c("Middle", "Poorest", "Poor", "Rich", "Richest")
index <- c(2, 3, 1, 4, 5)
winames[index]

## [1] "Poorest" "Poor"    "Middle"  "Rich"    "Richest"

Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
Xs <- matrix(rep(Xs, 5), nrow = 5, byrow = T)
Xs[,10] <- c(0, 1, 0, 0, 0)
Xs[,11] <- c(0, 0, 1, 0, 0)
Xs[,12] <- c(0, 0, 0, 1, 0)
Xs[,13] <- c(0, 0, 0, 0, 1)

eXB.post <- exp(t(Xs*%THETA.post.mat) )
qE <- apply( eXB.post, 2, quantile, probs=c(.025, .5, .975))

sample_size <- sum(wm$windex5==3)
```

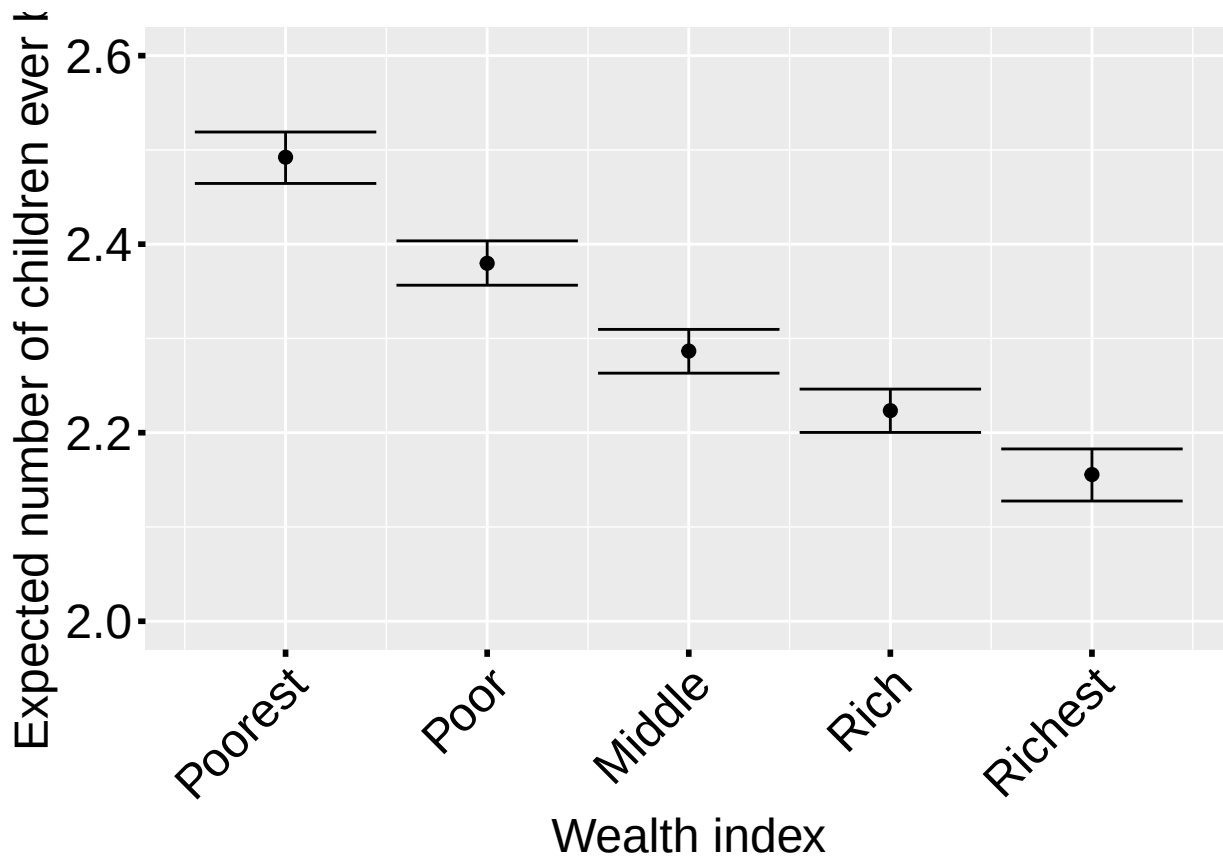
```

for (wealth in c(1,2,4,5)){
  sample_size <- c(sample_size, sum(wm$windex5==wealth))
}
qE <- rbind(qE,sample_size)

colnames(qE)<-winames
write.csv(t(qE), paste(IMAGEOUT,"qE.windex.csv",sep=""))

BP.df_windex5 <- data.frame(x =1:5,
  F = c(qE[2,])[index],
  L = c(qE[1,])[index],
  U = c(qE[3,])[index])
abc <- ggplot(BP.df_windex5, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Wealth index", cex.lab = 2.5, cex.axis = 1.5) +
  geom_point(size = 2) +
  ylim(2,2.6) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:5), labels = winames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
  axis.ticks.x = element_line(color = "black", linewidth = 1),
  axis.text.y = element_text(size = 18, color = "black"),
  axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)

```



```

save(BP.df_windex5, file=paste(IMAGEOUT,"BP.df_windex5.RData",sep=""))
dev.print(pdf, paste(IMAGEOUT, "E_windex5.pdf",sep=""),width=8,height=8)

```

```
## pdf
## 2
```

Expected Children Ever Born - other wives

```
anames<- c("No", "Yes")
index <- c(1, 2)
anames[index]
```

```
## [1] "No" "Yes"
```

```
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
Xs <- matrix(rep(Xs, 2), nrow = 2, byrow = T)
Xs[,15] <- c(1, 2) # 2 for Yes, 1 for No
```

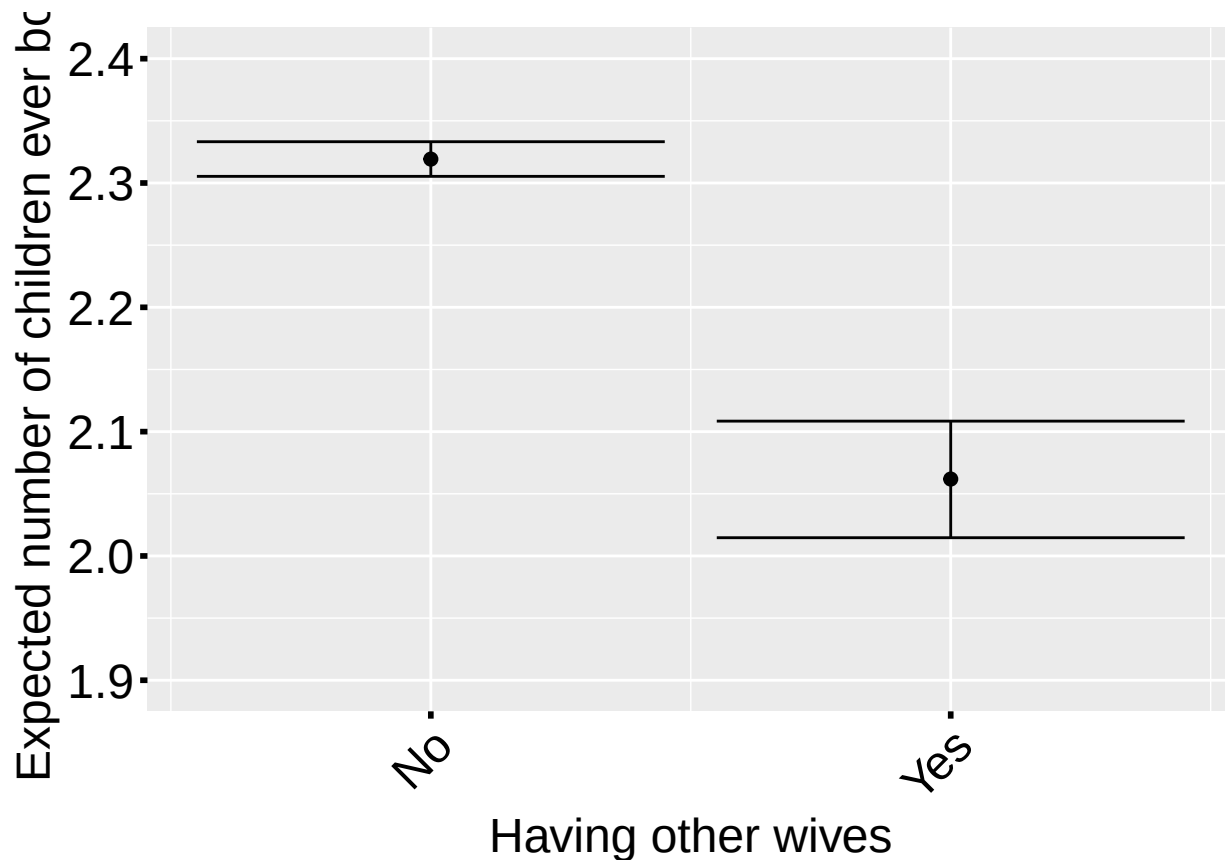
```
eXB.post<- exp(t(Xs%%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
```

```
sample_size <- c(sum(X[,14]==1), sum(X[,14]==2)) # No, Yes
qE <- rbind(qE,sample_size)
```

```
colnames(qE)<-anames
write.csv(t(qE), paste(IMAGEOUT,"qE.other_wives.csv",sep=""))
```

```
BP.df_other_wives <- data.frame(x =1:2,
                                F = c(qE[2,])[index],
                                L = c(qE[1,])[index],
                                U = c(qE[3,])[index])
```

```
abc <- ggplot(BP.df_other_wives, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Having other wives") +
  geom_point(size = 2) +
  ylim(1.9,2.4) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:2), labels = anames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)
```



```
save(BP.df_other_wives, file=paste(IMAGEOUT,"BP.df_other_wives.RData",sep=""))
dev.print(pdf, paste(IMAGEOUT, "E_other_wives.pdf",sep=""),width=8,height=8)
```

```
## pdf
## 2
```

```
(qE[2,][1] - qE[2,][2])/qE[2,][2]
```

```
## No
## 0.124824
```

Expected Children Ever Born - media

```
anames<- c("No", "Yes")
index <- c(1, 2)
anames[index]
```

```
## [1] "No" "Yes"
```

```
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
Xs <- matrix(rep(Xs, 2), nrow = 2, byrow = T)
Xs[,14] <- c(1, 2) # 2 for Yes, 1 for No
```

```
eXB.post<- exp(t(Xs*%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
```

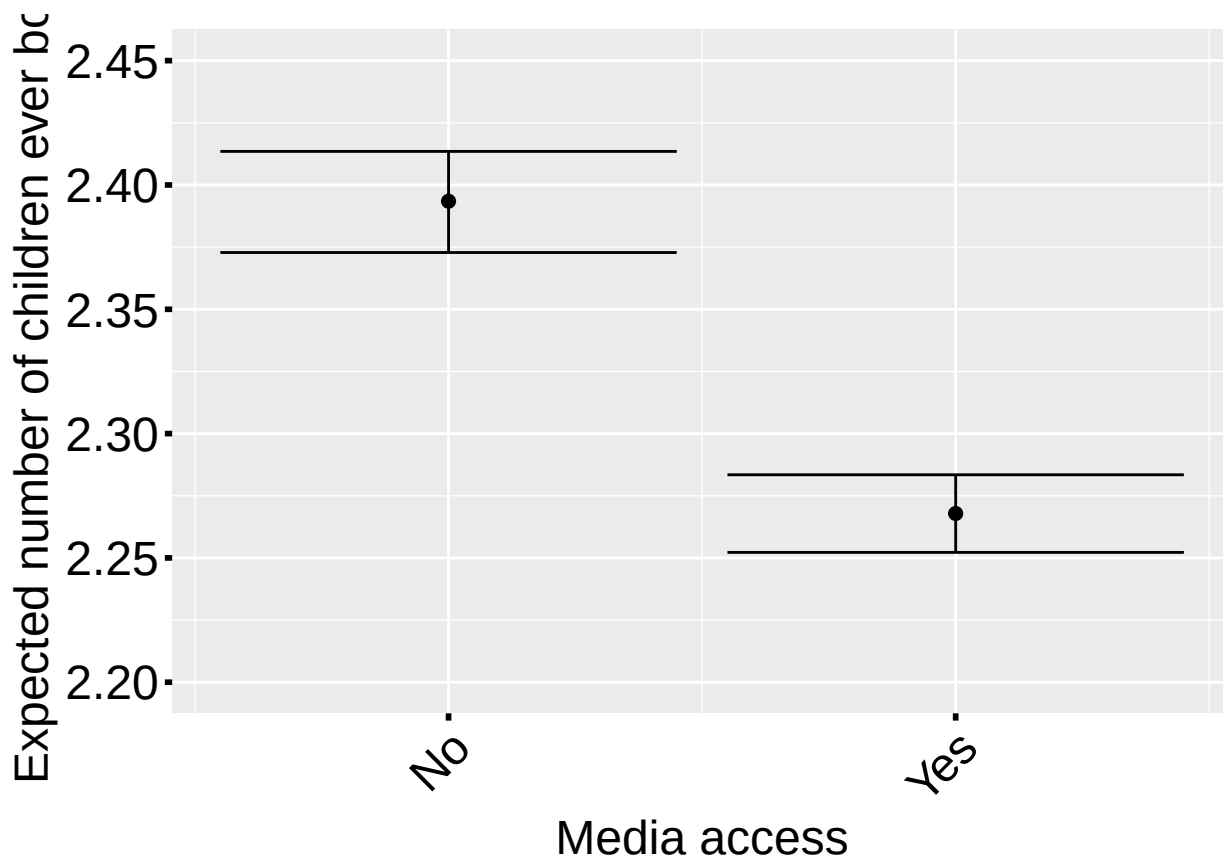
```

sample_size <- c(sum(X[,13]==1), sum(X[,13]==2)) # No, Yes
qE <- rbind(qE,sample_size)

colnames(qE)<-anames
write.csv(t(qE), paste(IMAGEOUT,"qE.media.csv",sep=""))

BP.df_media <- data.frame(x = 1:2,
                          F = c(qE[2,])[index],
                          L = c(qE[1,])[index],
                          U = c(qE[3,])[index])
abc <- ggplot(BP.df_media, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Media access") +
  geom_point(size = 2) +
  ylim(2.2,2.45) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:2), labels = anames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)

```



```

save(BP.df_media, file=paste(IMAGEOUT,"BP.df_media.RData",sep=""))
dev.print(pdf, paste(IMAGEOUT, "E_media.pdf",sep=""),width=8,height=8)

```

```

## pdf
## 2

```

```
(qE[2,][1] - qE[2,][2])/qE[2,][2]
```

```
##           No  
## 0.05537851
```

Expected Children Ever Born - Household head's sex

```
anames<- c("Male", "Female")  
index <- c(1, 2)  
anames[index]
```

```
## [1] "Male"   "Female"
```

```
Xs <- apply(X, 2, mean)  
Xs[2] <- Xs[1]^2  
Xs <- as.matrix(c(1, Xs))  
Xs <- matrix(rep(Xs, 2), nrow = 2, byrow = T)  
Xs[,17] <- c(1, 2) # 2 for Female, 1 for Male
```

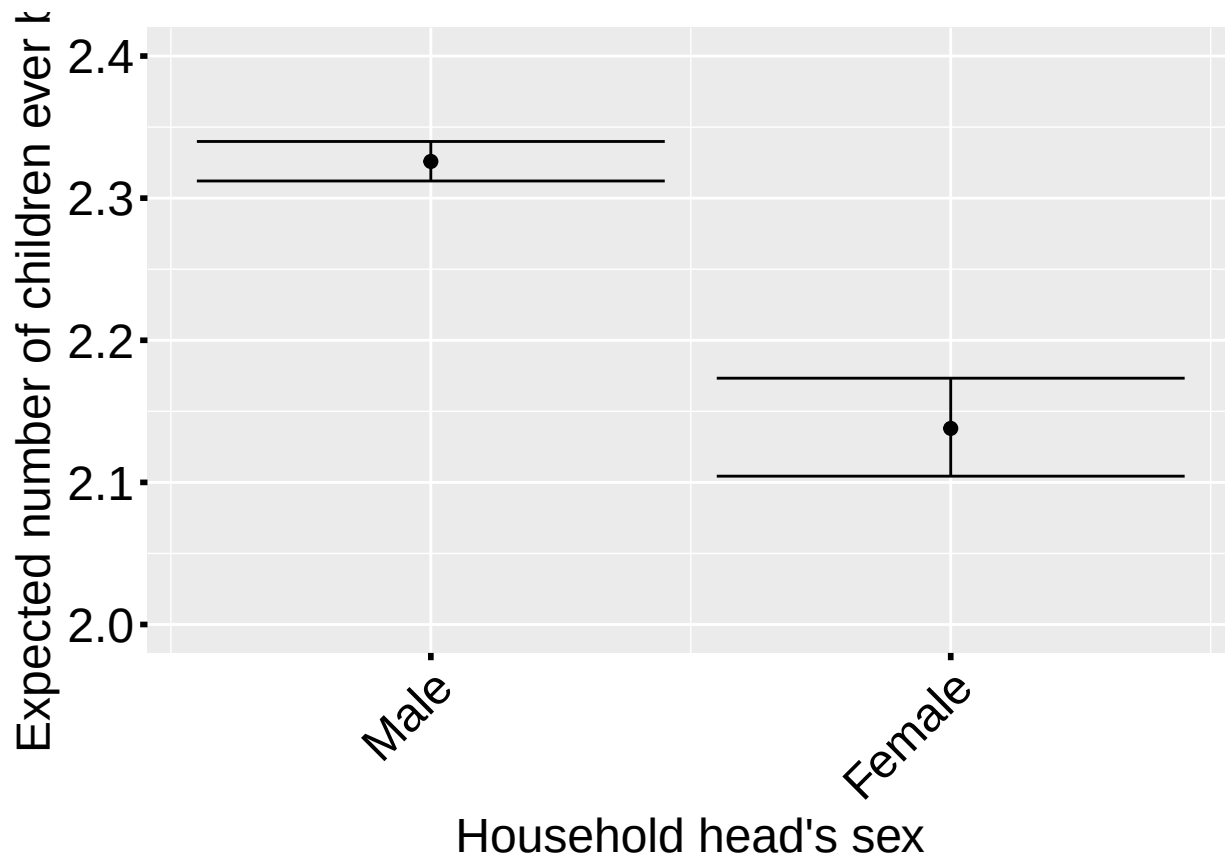
```
eXB.post<- exp(t(Xs%*%THETA.post.mat) )  
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
```

```
sample_size <- c(sum(X[,16]==1), sum(X[,16]==2)) # Male, Female  
qE <- rbind(qE,sample_size)
```

```
colnames(qE)<-anames  
write.csv(t(qE), paste(IMAGEOUT,"qE.HH_sex.csv",sep=""))
```

```
BP.df_HH_sex <- data.frame(x =1:2,  
                           F = c(qE[2,])[index],  
                           L = c(qE[1,])[index],  
                           U = c(qE[3,])[index])
```

```
abc <- ggplot(BP.df_HH_sex, aes(x = x, y = F)) +  
  labs(y = "Expected number of children ever born", x = "Household head's sex") +  
  geom_point(size = 2) +  
  ylim(2.0,2.4) +  
  geom_errorbar(aes(ymax = U, ymin = L)) +  
  scale_x_continuous(breaks = c(1:2), labels = anames[index]) +  
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 0),  
        axis.ticks.x = element_line(color = "black", linewidth = 1),  
        axis.text.y = element_text(size = 18, color = "black"),  
        axis.ticks.y = element_line(color = "black", linewidth = 1))  
print(abc)
```



```
save(BP.df_HH_sex, file=paste(IMAGEOUT,"BP.df_HH_sex.RData",sep=""))
dev.print(pdf, paste(IMAGEOUT, "E_HH_sex.pdf",sep=""),width=8,height=8)
```

```
## pdf
## 2
```

```
(qE[2,][1] - qE[2,][2])/qE[2,][2]
```

```
## Male
## 0.08785504
```

DIC Calculation

```
N_THIN=(S-B)/(thin)
DIC_array<-array(dim=N_THIN)
for(i in 1:N_THIN)
{
  DIC_i <- sum(-2 * dpois(y, lambda=exp(matmul(X,as.matrix(THETA.post.mat[,i])) ),log=TRUE))
  DIC_array[i] <- DIC_i
}
```

```
DIC_bar = mean(DIC_array)
D_mean <- sum(-2 * dpois(y, lambda=exp(matmul(X,as.matrix(apply(THETA.post.mat, 1, mean))) ),log=TRUE))
DIC <- 2*DIC_bar - D_mean
# 111691
```